# London dwelling-stock change, 2012-2021: summary of findings

This summarises what our model of annual dwelling-stock change across London's LSOAs currently shows. For a browsable, area-by-area version of everything below, see the live dashboard: dwelling-change-dashboard.

## Total change across London, 2011–2021

Between the 2011 and 2021 Censuses, London's dwelling stock grew by an estimated **~354,000 homes**. This is the Census-recorded figure itself, not a separate model estimate of it.

For this round of modelling, we have chosen to treat that Census figure — for London overall and for every borough and small area within it — as exactly correct. This is a deliberate simplifying assumption, not a claim that the Census itself is free of error: Censuses are large, careful exercises but are not perfect, and it is possible the true figure differs somewhat from the recorded one. We are choosing not to model that possibility in this round, because doing so would add real complexity for a question (year-by-year timing) where it is unlikely to change the picture much. It is a choice we could revisit in future work if there were a specific reason to think Census error mattered here — treating the total as uncertain too would let that uncertainty flow through to every number below.

Given that starting assumption, the model adds no further uncertainty of its own to the decade total for London or for any borough: it is built so that every area's yearly figures always add up, by construction, to that area's Census-recorded 10-year total.

The genuine modelling question — and the one this whole exercise is about — is **how that change was distributed year by year**. That is where the model's own uncertainty lives, on top of whatever uncertainty already exists in the Census figure itself, and it varies a lot from area to area.

For example, London's estimated change by year (with a 90% plausible range):

| Year | Estimated change | Plausible range |
|---|---:|---:|
| 2012 | 32,100 | 30,700 – 33,600 |
| 2013 | 34,100 | 32,900 – 35,400 |
| 2014 | 34,900 | 33,700 – 36,200 |
| 2015 | 38,000 | 36,700 – 39,200 |
| 2016 | 44,400 | 42,900 – 45,900 |

| Year | Estimated change | Plausible range |
|---|---:|---:|
| 2017 | 39,600 | 38,400 – 40,800 |
| 2018 | 36,400 | 35,100 – 37,700 |
| 2019 | 34,700 | 33,300 – 36,100 |
| 2020 | 32,200 | 31,000 – 33,400 |
| 2021 | 27,500 | 26,100 – 28,800 |

## Confidence by area

We looked at all 4,987 London LSOAs individually. Each one falls into one of three groups:

- **Confident (47% of areas)** — we have a single, reliable year-by-year picture of when the change happened.
- **A small number of likely stories (29% of areas)** — we can't pin down exactly which year(s) the change happened in, but the uncertainty narrows down to 2 or 3 specific, describable possibilities, each with a stated likelihood (e.g. "60% likely 2019, 40% likely 2021").
- **Genuinely unclear (24% of areas)** — the total change for the area is just as reliable as anywhere else, but we cannot say with any confidence which year(s) it happened in. This is usually because the underlying planning/OS AddressBase records for that area are too sparse or too inconsistent over the decade to distinguish between years. We report the total only in these cases, rather than guessing.

We report this last group directly rather than omitting it. It is a real, sizeable minority, and it reflects a genuine limit of the underlying records, not a shortcoming we expect to fix by refining the model further.

### Example areas

- **A confident area** — an LSOA in Newham grew by about 720 homes over the decade, and we're confident it happened in four clear bursts (around 2015, 2016, 2018 and 2020), with next to nothing in the other six years.
- **A "small number of stories" area** — an LSOA in Islington grew by about 480 homes. We can't say for certain which year it happened in, but the possibilities narrow to two: roughly 63% likely it was 2021, 37% likely it was 2019.
- **A genuinely unclear area** — an LSOA in Tower Hamlets grew by about 380 homes, but the records don't let us say in which year(s) — only that it happened sometime across the decade.

## Variation by borough

Growth is concentrated in a fairly small number of boroughs: **Tower Hamlets** (~29,000 homes), **Newham** (~22,000), **Southwark** (~19,000), **Greenwich** (~16,000) and **Barnet** (~16,000) account for a disproportionate share of London's total growth. At the other end, **City of London**, **Richmond upon Thames** and **Kingston upon Thames** each saw only a few thousand.

Confidence in the year-by-year picture also varies by borough — generally, the areas where we're least sure *when* the change happened are the same ones with the most activity to reconcile (more growth means more records to cross-check, and more opportunity for them to disagree). Richmond upon Thames, Enfield and Kingston upon Thames have the most areas with a confident year-by-year picture; City of London, Tower Hamlets and Bexley have the fewest — for City of London and Tower Hamlets, that's a direct consequence of having some of the highest volumes of change to account for.

## Open questions and next steps

- **Treating the Census counts as exact is a choice for this round, not a settled fact.** If a future need arose to account for possible Census error too, that would add genuine uncertainty to the total figures above (currently shown as fixed) as well as to the year-by-year ones — a bigger piece of work than anything in this round, and not something we recommend doing without a specific reason to.
- **The 24% "genuinely unclear" group is a property of the underlying records for those specific areas, not something we expect further modelling work to resolve.** If more granular or more consistent planning or OS AddressBase-derived data becomes available for those areas in future, this could improve — but that's a data question, not a modelling one.
- **Both of our two main data sources — planning completions data and the OS AddressBase-derived dwelling-change records — likely have their own data-quality issues that we have not yet fully characterised.** Investigating both sources directly is a next step already underway, separate from this modelling work. Separately, if there is existing expert knowledge of specific known issues with either source (e.g. particular boroughs, years, or record types that are known to be unreliable), that knowledge can be built into a future version of the model directly — this is a case where local expertise could meaningfully improve the estimates, and we'd welcome it.
- **This kind of model can also be extended to draw on other data sources** — Energy Performance Certificate (EPC) records are one candidate that could plausibly help distinguish which years changes happened in, if there's an appetite to explore adding a third data source alongside planning and OS AddressBase data.
- **We have tested, but not included, two other modelling ideas**: accounting for a delay between when a change happens and when it shows up in the records ("temporal lag"), and letting changes in one area inform the picture in neighbouring areas ("spatial spillover"). Neither showed a clear improvement over the simpler model reported here once tested properly — the spatial-spillover version scored clearly worse on our most rigorous accuracy check, and

the lag version, combined with this model, gave at best a marginal improvement on one measure while making the year-by-year confidence picture worse. We haven't ruled out a different way of building in either idea helping in future, but based on what we've tried so far we don't expect adding them back in to meaningfully change the totals reported above.

- **The dashboard is the right tool for area-specific questions** — anyone can look up a specific LSOA or borough and see exactly which group it falls into and why, rather than relying on the borough-level averages above. It also breaks down which kinds of areas tend to fall into the "genuinely unclear" group — in short, areas with a larger amount of change to account for are systematically less likely to have a confident year-by-year picture, regardless of borough.
- Questions or requests for a specific area/borough breakdown not covered here should go to the modelling team directly.